

=== CONFIGURACION GNB (1780306191) ===

```
# gnb.yaml
# srsRAN 5G gNB configuration for URLLC
#
# This configuration is optimized for low latency with a 30kHz SCS
# and uses ZMQ for communication with the UE.

# Core network configuration
amf_addr: 10.53.1.2
amf_port: 38412
bind_addr: 10.53.1.3
next_hop_s_nssai:
  sst: 1
  sd: 1

# gNB parameters
mcc: 315
mnc: 010
nci: 0x000001
id: 0x000001
tac: 1

# Radio Frequency (RF) configuration
[rf]
srate: 30.72e6
tx_gain: 70
rx_gain: 30
nof_antennas: 1

# NR RAT configuration
[rat.nr]
bands: 78
nof_carriers: 1
max_nof_prb: 106
nof_prb: 106
dl_nr_arfcn: 700000
ssb_nr_arfcn: 700010
common_scs: 30
ssb_scs: 30

# RRC configuration
[rrc]
release: 15
slice:
  - sst: 1
    sd: 1

# PCAP logging (disabled by default)
[pcap]
enable: none
mac_filename: /tmp/gnb_mac.pcap
mac_nr_filename: /tmp/gnb_mac_nr.pcap
nas_filename: /tmp/gnb_nas.pcap

# Logging configuration
[log]
all_level: info
phy_lib_level: none
all_hex_limit: 32
filename: /tmp/gnb.log
file_max_size: -1

# Device configuration (ZMQ for UE communication)
[device]
device_name: zmq
```

```
device_args: "driver=zmq,tx_port=2152,rx_port=2153,base_srate=30.72e6"
```

```
# CU-CP configuration
```

```
[cu_cp]
```

```
bind_addr: 10.53.1.3
```

=== CONFIGURACION UE (1780306191) ===

```
# ue.conf
# srsRAN 5G UE configuration for URLLC
#
# This configuration is optimized for low latency with a 30kHz SCS
# and uses ZMQ to communicate with the gNB.

# Radio Frequency (RF) configuration
[rf]
freq_offset: 0
tx_gain: 10
rx_gain: 5
srate: 30.72e6
nof_antennas: 1

# NR RAT configuration
[rat.nr]
bands: 78
nof_carriers: 1
max_nof_prb: 106
nof_prb: 106
dl_nr_arfcn: 700000
ssb_nr_arfcn: 700010

# PCAP logging (disabled by default)
[pcap]
enable: none
mac_filename: /tmp/ue_mac.pcap
mac_nr_filename: /tmp/ue_mac_nr.pcap
nas_filename: /tmp/ue_nas.pcap

# Logging configuration
[log]
all_level: info
phy_lib_level: none
all_hex_limit: 32
filename: /tmp/ue.log
file_max_size: -1

# USIM configuration
[usim]
mode: soft
algo: milenage
opc: 63BFA50EE6523365FF14C1F45F88737D
k: 00112233445566778899aabbccddeeff
imsi: 001010123456780
imei: 353490069873319

# RRC configuration
[rrc]
release: 15
ue_category: 4

# NAS configuration
[nas]
apn: srsapn
apn_protocol: ipv4

# Gateway configuration (no IP address defined as per instructions)
[gw]
# ip_addr is intentionally omitted as per instructions.

# GUI configuration
[gui]
enable: true
```

```
# Device configuration (ZMQ for gNB communication)
```

```
[device]
```

```
device_name: zmq
```

```
device_args: "tx_port=2153,rx_port=2152,master_clock_rate=30.72e6,base_srate=30.72e6,bind_addr=10.53.1.3"
```

=== DOCKER COMPOSE (1780306191) ===

```
# docker-compose.yml
# Docker Compose configuration for Open5GS Core and srsRAN gNB
#
# This setup creates a dedicated network for the 5G core and gNB,
# assigning static IPs for reliable communication.

version: '3.8'

networks:
  srsran-net:
    driver: bridge
    ipam:
      config:
        - subnet: 10.53.1.0/24

services:
  # Open5GS Core Network Services
  5gc:
    container_name: 5gc
    image: open5gs/open5gs:latest
    privileged: true
    environment:
      - TZ=Europe/Madrid
    volumes:
      - ./open5gs/etc:/open5gs/etc
      - ./open5gs/db:/open5gs/db
    command: ["/bin/bash", "-c", "for service in amf smf upf ausf pcf nssf udr udm webui; do open5gs-$service & done &"]
  networks:
    srsran-net:
      ipv4_address: 10.53.1.2
  ports:
    - "3000:3000" # WebUI
    - "38412:38412/sctp" # AMF N2
    - "2152:2152/udp" # UPF N3
    - "2153:2153/udp" # UPF N3
  cap_add:
    - NET_ADMIN
    - SYS_NICE
  healthcheck:
    test: ["CMD-SHELL", "pgrep open5gs-amf || exit 1"]
    interval: 10s
    timeout: 5s
    retries: 5
  restart: unless-stopped

# srsRAN gNB
gnb:
  container_name: gnb
  image: srsran_project/srsran-docker-build:latest
  privileged: true
  environment:
    - TZ=Europe/Madrid
  volumes:
    - ./gnb.yaml:/etc/srsran/gnb.yaml
  command: ["/bin/bash", "-c", "srsgnb -c /etc/srsran/gnb.yaml"]
  networks:
    srsran-net:
      ipv4_address: 10.53.1.3
  ports:
    - "2152:2152/udp" # ZMQ tx_port for gNB (UE rx_port)
    - "2153:2153/udp" # ZMQ rx_port for gNB (UE tx_port)
  cap_add:
    - NET_ADMIN
    - SYS_NICE
```

```
healthcheck:  
  test: ["CMD-SHELL", "pgrep srsgnb || exit 1"]  
  interval: 10s  
  timeout: 5s  
  retries: 5  
restart: unless-stopped
```